12.2 The Base-Rate Fallacy

WORKSHEET

The question

STEP II 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Imagine 1000 people. How many have the disease? How many of them test positive? How many of the healthy people test positive?

STAGE 1 Bayes' theorem

Ask yourself Write D for "Frankie has the disease" and $+$ for "AL is positive". You know $\mathrm{P}(+ \mid D)$ and want $\mathrm{P}(D \mid +)$. How are they related?

STAGE 2 A second test

Ask yourself The second test is given only to those who tested positive on AL. What is the proportion of sufferers in that group? Use it as the new "base rate".

Pause and try With $p = 0.09$, STEP's false-positive rate is about the same as AL's, yet a second positive result raises the probability from $\frac{1}{11}$ to $\frac{1}{2}$. Why does the same quality of test have a much bigger effect the second time?

What's really going on

Odds and evidence

Bayes' theorem is clearest in terms of *odds*. The odds of D are $\mathrm{P}(D) : \mathrm{P}(\bar{D})$. A piece of evidence E multiplies the odds by the *likelihood ratio* $\frac{\mathrm{P}(E|D)}{\mathrm{P}(E|\bar{D})}$:

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}.$$

Here the prior odds are $1 : 99$, and AL's likelihood ratio is $0.9 \div \frac{1}{11} = 9.9$, giving posterior odds $9.9 : 99 = 1 : 10$, that is, probability $\frac{1}{11}$. A second independent test with likelihood ratio 10 (as STEP with $p = 0.09$ has) brings the odds to $1 : 1$. Taking logarithms, evidence *adds*: each independent test contributes the logarithm of its likelihood ratio. Alan Turing used exactly this, measuring evidence in "bans" and "decibans", to break Enigma messages at Bletchley Park.

The prosecutor's fallacy

In 1999 the solicitor Sally Clark was convicted of murdering her two baby sons, partly on the strength of evidence that the chance of two cot deaths in one family was "1 in 73 million". Quite apart from the error in that number (it assumed the two deaths were independent, ignoring genetic and environmental factors), the reasoning confused $\mathrm{P}(\text{evidence} \mid \text{innocent})$ with $\mathrm{P}(\text{innocent} \mid \text{evidence})$. Double murder by a mother is also extremely rare, and the relevant comparison is between the two explanations. The Royal Statistical Society wrote publicly to point out the error, and her conviction was overturned in 2003. The mathematics is exactly that of this question.

Gigerenzer and natural frequencies

In the 1990s the psychologist Gerd Gigerenzer showed that doctors who were given test statistics as conditional probabilities mostly got questions like this one badly wrong, but that when the same information was presented as natural frequencies

("of 1000 women, 10 have breast cancer; of these, 8 test positive…") most got them right. His work has changed how medical risks are communicated to patients in many countries.

Try this next

1. **A body in the fridge.** Try STEP II 1987, Question 16 (step.maths.org/questions/87-s2-q16): Bayes' theorem with two witnesses of different reliability, in one of STEP's more memorable settings.

2. **How rare?** For a test with sensitivity 0.9 and false-positive rate $\frac{1}{11}$, how common must a disease be for a positive result to mean a better than even chance of having it?

3. **Repeat the test.** If AL itself is repeated, and the results are independent given the person's true state, what is the probability of disease after two positive AL results? After three?

4. **Screening.** A screening programme tests a million people for a condition with prevalence 1 in 10 000, using a test with sensitivity 99% and false-positive rate 1%. How many people will be told they may have the condition, and how many of them actually do?

Further reading

- **Gerd Gigerenzer, *Reckoning with Risk*** (Penguin). Natural frequencies, medical tests and the courtroom.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican). Includes medical screening and the Sally Clark case.
- **Sharon Bertsch McGrayne, *The Theory That Would Not Die*** (Yale University Press). The history of Bayes' theorem, including Turing's use of it.

The full guided solution is in Section 12.2:
https://beyondstep.keithmcnulty.org/ch12/base-rate.html